

Training-Free RL on Frozen Omni Speech Models: Reward-Guided Best-of- N , a Paralinguistic Probe, and a Reward-Spread Lens

Exploring-L4-Intelligence Project

Abstract

We study *training-free reinforcement learning on frozen omni speech models*: reward-guided optimization applied purely at inference time, with no weight updates and no architecture change. Our primary result is a *genuine* reward-driven best-of- N on a frozen *generative* omni model. On Qwen3-Omni-30B-A3B (Q8_0 GGUF, driven by llama.cpp), held fixed, we sample N candidate transcripts per utterance and select with a verifiable reward—best-of- N over model-sampled candidates, the concrete realisation of Gibbs tilting $q^* \propto q_0 \exp(R/\beta)$. On LibriSpeech test-other with additive noise (snr5), the verifiable-reward argmax (the $\beta \rightarrow 0$ tilt) cuts word error rate from a greedy 0.117 to 0.073 at $N = 8$, a reduction of +0.044 (95% paired-bootstrap CI [0.024, 0.067], significant) whose headroom *scales with N* (+0.013, +0.032, +0.044 for $N = 2, 4, 8$); the deployable, label-free MBR-consensus selector realises +0.014 (95% CI [-0.0001, 0.033], boundary non-significant). The reward-driven headroom is thus real, significant, and N -scaling on a frozen model, while the deployable selector captures only a small fraction of it—an honest realized-versus-headroom gap. We read this through a lightweight lens: the gain of selection is $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$, bounded by the *reward spread* over the model’s sampled candidates, the Donsker–Varadhan / free-energy identity used *qualitatively*, with two strict sorry-free Lean 4 lemmas (a flat reward yields zero gain, `flat_no_gain`; a non-constant reward yields strictly positive gain, `gain_pos_of_nonconstant`). Separately, we *probe*—not RL—a frozen content bi-encoder (omni-embed-nemotron-3b): its embeddings carry near-chance speaker identity (mean acc 0.033, $3.0\times$ chance, 91-way) and only modest emotion (0.405, $2.4\times$ chance, 6-way) on CREMA-D, and training-free pool selection yields no significant emotion gain (95% CI [-0.043, +0.116]). We state the caveats plainly: llama.cpp audio input is experimental upstream; snr5 is a robustness perturbation chosen for measurable headroom; the oracle uses the reference and is an upper bound, not deployable. All results are on frozen models with committed reproduction artifacts.

1 Introduction

We ask how much can be extracted from *frozen omni speech models* without touching a weight or changing their structure, using reward-guided optimization applied purely at inference time. A modern omni or speech-capable multimodal LLM is, by release, an extraordinary repository of latent capability: from large-scale audio, parallel speech-text corpora, and instruction mixtures, models such as Qwen2-Audio, Qwen2.5-Omni, Qwen3-Omni, and SALMONN [qwe, 2024, 2025a,b, sal, 2023] acquire transcription, translation, speaker identity, paralinguistic affect, and dialogue competence simultaneously. Our question is deliberately conservative: *how far can one go with a model held fixed?* We treat “training-free RL” concretely as inference-time *verifiable-reward selection*—above all best-of- N over candidates the model itself samples—that changes no parameters and no architecture [bei, 2024, mud, 2023, zuo, 2025]. Our primary result realises this genuinely: a frozen *generative* omni model produces N transcript candidates per utterance, and a verifiable reward selects among them. A second study is honest *probing* of a frozen content encoder—a distinct operator that we are careful *not* to dress up as reward-driven selection. Every result below is on a frozen model; no weights change anywhere.

The lever is the reward spread. The premise that organizes this paper is that the gain available to inference-time selection is a property of *which reward, over which candidates* one selects. When the reward is nearly *flat* over the candidate set the frozen model actually places mass on, selection moves through a narrow envelope of near-identical rewards and cannot manufacture headroom the candidate distribution does

not contain. When the model’s samples *spread* in reward, selection can climb, and the headroom grows as more candidates are drawn. For speech in particular, a generative omni model under acoustic stress samples transcripts of genuinely different quality—so a verifiable transcription reward has spread to exploit—whereas a frozen content encoder tends to “read” a transcribable surface rather than “listen” to the acoustic substrate, exposing spread for content and intent but little for paralinguistics [min, 2022, vox, 2026, hsu, 2023]. The central object of the paper is this reward *spread*, and the empirical fact that it is large for the model’s noisy transcript candidates and small for paralinguistic read-out.

Selection as Gibbs tilting. To reason about this quantitatively we adopt the standard variational view of inference-time alignment. Optimizing the KL-regularized objective

$$F(q) = \mathbb{E}_{z \sim q}[R(z)] - \beta \text{KL}(q \parallel q_0)$$

over selection distributions q on the candidate space $\mathcal{Z}$ has the closed-form Gibbs optimum $q^*(z) \propto q_0(z) \exp(R(z)/\beta)$, with partition function Z [kor, 2022, bei, 2024]. Best-of- N and its soft refinements [ver, 2025, yan, 2024, gui, 2024], self-consistency [wan, 2022], and MBR/reranking [ich, 2025, mbr, 2025] are all estimators of this tilt over some realization of $\mathcal{Z}$. Taking q_0 to be the frozen generative model’s sampling distribution, the verifiable-reward argmax is the $\beta \rightarrow 0$ limit of the tilt, and its attainable benefit is bounded by how far q^* can depart from q_0 —which is governed by the spread of R over $\text{supp}(q_0)$, i.e. over the candidates the model actually draws.

A reward-spread lens. Our lens is a single, lightweight identity, stated at its true strength rather than as a discovery. Granted the Gibbs framing, the gain is exactly $\mathcal{G} = F(q^*) - F(q_0) = \beta \text{KL}(q_0 \parallel q^*)$, and a Hoeffding argument caps it by the reward spread, $\mathcal{G} \leq \text{spread}^2/(8\beta)$; to leading order the discriminator sharpens to $\mathcal{G} \approx \text{Var}_{q_0}(R)/(2\beta)$. This is the Donsker–Varadhan / free-energy identity, and we are explicit that we use it *qualitatively*, not as a novel theorem. Its two load-bearing corners are machine-checked sorry-free in Lean 4: a flat reward yields zero gain (`flat_no_gain`), and a non-constant reward yields strictly positive gain (`gain_pos_of_nonconstant`). The reading is direct, and it now speaks about our primary experiment: flat reward across the model’s sampled transcripts $\Rightarrow$ no best-of- N headroom; spread across them $\Rightarrow$ headroom that grows with N .

Relation to prior training-free work. Training-free selection—best-of- N , reranking, MBR—is established, most visibly for ASR and open-ended generation [bei, 2024, mbr, 2025, ich, 2025]. We do *not* claim novelty in the best-of- N mechanism itself. Our contribution is threefold and honestly demarcated: (i) a genuine reward-driven best-of- N run on a *frozen generative omni* model under acoustic stress, with a significant, N -scaling oracle headroom and an honest realized-versus-headroom gap for the deployable selector; (ii) a precise *probing* study of a frozen content encoder, reported as probing rather than RL; and (iii) the reward-spread lens that ties the two together. The encoder study and the generative best-of- N are *distinct operators*; we do not claim identical machinery between them.

Contributions. We make exactly three.

(C1) A genuine reward-driven best-of- N on a frozen generative omni model. On Qwen3-Omni-30B-A3B (Q8_0 GGUF via llama.cpp), held fixed, we sample N candidate transcripts per utterance and select with a verifiable reward: genuine best-of- N over model-sampled candidates. On LibriSpeech test-other with additive noise (snr5, chosen for measurable headroom; $n = 96$, pool = 8, temperature 0.8, seed 42; paired-bootstrap CIs on per-utterance WER reduction), greedy decoding gives WER 0.117. The oracle-WER best-of- N —the verifiable-reward argmax, the $\beta \rightarrow 0$ tilt—reduces WER to 0.073 at $N = 8$, a reduction of +0.044 (95% CI [0.024, 0.067], significant), and the headroom *scales with N* (+0.013, +0.032, +0.044 for $N = 2, 4, 8$)—the empirical best-of- N shape. The deployable, reference-free MBR-consensus selector (which uses no labels) reaches WER 0.102, +0.014 (95% CI [−0.0001, 0.033], boundary non-significant). The verifiable-reward headroom is real, significant, and N -scaling on a frozen model; the label-free deployable selector realises only a small, boundary-significant fraction of it. Committed artifact `_repro/asr_bon_llamacpp_snr5.json`, with the reproduction command committed. Caveats stated plainly: llama.cpp audio input is experimental

upstream (“may have reduced quality”); snr5 additive noise is a robustness perturbation chosen to expose headroom; the oracle uses the reference (an upper bound, not deployable), while MBR is the deployable proxy.

(C2) A frozen-encoder probe (not RL, a distinct operator). We *probe*—we do not run reward-driven selection on—a frozen content bi-encoder, omni-embed-nemotron-3b. On CREMA-D (dev-selected layer, 3 seeds; committed `_repro/paralinguistic_negative_probe.json`) its frozen embeddings carry near-chance speaker identity (mean acc 0.033, $3.0\times$ chance, 91-way) and only modest emotion (0.405, $2.4\times$ chance, 6-way); training-free pool selection yields *no* significant emotion gain (across-seed 95% CI $[-0.043, +0.116]$, spanning zero; committed `_repro/emotion_pool_paired_v2.json`). Honestly noted, and *not* headlined as a selection gain: the same encoder does zero-shot SLU/intent classification by candidate-card cosine retrieval (MInDS-14 en-US Acc@1 = 0.984; committed `_repro/minds14_toolintent_paired.json`). This is a card-representation / zero-shot-classification result—its verifiable reward is only the *evaluation* metric and does *not* drive selection—and it is a distinct operator from C1’s generative best-of- N . Any card comparison is reported plainly as a zero-shot-classification ablation whose card-design gain is confounded (in-set example cards plus a query-instruction change), not as a training-free-RL selection gain.

(C3) A lightweight reward-spread lens. We formalize the gain identity $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$ and its spread ceiling, use it qualitatively as the Donsker–Varadhan / free-energy identity, and support it with two strict, sorry-free Lean 4 lemmas (`flat_no_gain`, `gain_pos_of_nonconstant`). The lens now genuinely explains C1: the oracle best-of- N headroom is governed by the spread of the verifiable reward across the model’s sampled candidates—flat reward across candidates yields no headroom, spread yields headroom that grows with N . It also explains the C2 contrast qualitatively: the frozen content encoder exposes reward spread for content/intent but little for paralinguistics.

Scope and honest framing. C1 is a genuine but *modest* training-free RL result: a significant, N -scaling oracle headroom, but a boundary-non-significant deployable selector, on an experimental audio path under a chosen robustness perturbation. C2 is honest probing of a frozen encoder, not RL, and a distinct operator. C3 is a lens, not a discovery. All results are on frozen models with committed reproduction artifacts; we under-claim deliberately.

Organization. Section 1 has framed the reward-spread lens and the three contributions. The remainder develops the Gibbs-tilt objective and the spread identity, with the two sorry-free Lean lemmas; presents the frozen generative best-of- N setup and the LibriSpeech noisy-ASR results (C1); reports the frozen-encoder probe—the CREMA-D paralinguistic negative and the zero-shot SLU note (C2); and closes with related work, limitations, and future directions.

2 Preliminaries and Problem Formulation

We study *training-free* reinforcement learning (RL): a regime in which a *frozen* multimodal speech/omni model is steered at inference time by a reward signal, with no gradient ever propagated into its parameters and no architectural surgery. Two distinct inference-time operators run on two distinct frozen omni speech models in this paper, and it is essential to keep them apart. The *primary* operator is *generative best-of- N* : from a frozen generative omni model we sample N candidate outputs and select among them by a verifiable reward. This is where the KL-regularized objective, its Gibbs optimum, and best-of- N genuinely apply—the candidates are model-sampled and the reward actually chooses one, so the procedure is a bona fide sampler from / estimator of the tilted target. The *secondary* operator is *frozen-encoder probing*: a frozen content bi-encoder is analyzed by embedding-cosine classification and linear/ k -NN probes. There the verifiable metric is an *evaluation* score, not a selector, and no Gibbs tilt over model-sampled candidates is performed; we say so plainly rather than dress the probe up as reward-driven selection. This section fixes notation, states the no-weight-update invariant, specifies both operators and their sharp boundary, introduces the reward-spread quantity that governs when selection helps, and specifies the verifiable reward class. The objects defined here are reused verbatim in the Theory section.

2.1 Training-free RL as constrained inference-time reward maximization

Let g_θ denote a pretrained speech/omni model with *frozen* parameters θ , and let x be an input (an utterance, a prompt, or a paired context). The inference-time degrees of freedom—the only quantities we are permitted to vary—are collected into an *action* $z \in \mathcal{Z}$, where $\mathcal{Z}$ is the selection action space made precise in Section 2.2. Executing an action produces an output that is scored by a bounded scalar *reward* $R: \mathcal{Z} \rightarrow [0, 1]$ (we write $R(z)$ for the reward of the output realized by z when x is fixed). A pretrained *base* (or prior) distribution $q_0 \in \Delta(\mathcal{Z})$ over actions is induced by the frozen model and the default inference procedure. Training-free RL searches over distributions $q \in \Delta(\mathcal{Z})$ for one that raises expected reward while remaining close to the prior.

Definition 1 (Training-free RL objective). Fix x , a base distribution q_0 , a reward R , and a temperature $\beta > 0$. The KL-regularized inference-time objective is

$$F(q) = \mathbb{E}_{z \sim q}[R(z)] - \beta \text{KL}(q \parallel q_0), \quad (1)$$

and the training-free RL problem is $q^* = \arg \max_{q \in \Delta(\mathcal{Z})} F(q)$.

The objective (1) is strictly concave in q over the simplex, and its maximizer is the classical Gibbs/exponential tilt of the prior [kor, 2022].

Proposition 1 (Gibbs optimum). *The unique maximizer of (1) is*

$$q^*(z) = \frac{1}{Z} q_0(z) \exp\left(\frac{1}{\beta} R(z)\right), \quad Z = \mathbb{E}_{z \sim q_0}\left[\exp\left(\frac{1}{\beta} R(z)\right)\right], \quad (2)$$

with optimal value $F(q^*) = \beta \log Z$.

Proof sketch. Adding a Lagrange multiplier for the normalization constraint and differentiating the concave functional (1) with respect to $q(z)$ yields $R(z) - \beta(\log q(z) - \log q_0(z)) - \beta = \text{const}$, i.e. $\log q(z) = \log q_0(z) + R(z)/\beta + \text{const}$; exponentiating and normalizing gives (2). Substituting back collapses the objective to $\beta \log Z$. $\square$

Equation (2) is the formal anchor of the paper’s *primary* result: the generative best-of- N procedure of Definition 3 is read as an *estimator of, or a sampler from, the tilted target q^** , and is therefore comparable through the lens of how well it approximates (2) at a given query budget N . The temperature β interpolates between the prior ($\beta \rightarrow \infty$, $q^* \rightarrow q_0$) and greedy reward maximization ($\beta \rightarrow 0^+$, $q^* \rightarrow$ the $\arg \max_z R$ point mass), recovering best-of- N selection in the low-temperature limit [bei, 2024, ver, 2025, kha, 2025]. The oracle best-of- N we report empirically is exactly this $\beta \rightarrow 0^+$ tilt over the model’s sampled candidates.

Definition 2 (No-weight-update invariant). A procedure is *training-free* if, for every input and every query budget, the model parameters θ and the model’s computational graph are held fixed; all adaptation is realized through the choice of q over the inference-time action space $\mathcal{Z}$. Formally, the map $\theta \mapsto \theta$ is the identity throughout, and only q is optimized.

Definition 2 is the operational contract of this work, and both operators below honor it: no weights change and no graph is edited on either frozen model. It excludes parameter-efficient fine-tuning and any backpropagation through g_θ . Because q^* depends on θ only through q_0 and on the task only through R , the frozen model’s pretrained competence is *activated*, never overwritten—the central premise the paper develops empirically and theoretically.

2.2 Two frozen-model operators: generative best-of- N and encoder probing

We instantiate two operators on two different frozen omni speech models. They share the no-weight-update invariant but are *not* the same machinery: only the first performs reward-driven selection over model-sampled candidates in the sense of (2).

Primary operator: generative best-of- N (this is the reward-driven selection). For a query x we draw a finite *candidate pool* $C(x) = \{y_1, \dots, y_N\}$ by sampling the frozen generative omni model g_θ at temperature τ ; the pool is genuinely produced by the model, not curated by us. The action space is the probability simplex over this pool, $\mathcal{Z} = \Delta(C(x))$; a *selection* action collapses to the vertex returning a single chosen candidate.

Definition 3 (Generative best-of- N selection operator). Let g_θ be a frozen generative omni model (in our primary experiments Qwen3-Omni-30B-A3B, an Q8_0 GGUF driven by llama.cpp). Sampling g_θ at temperature τ for a fixed budget N yields the pool $C(x)$ and the base distribution q_0 over it (the empirical sampling law of the model). A verifiable reward R scores each candidate, and training-free RL reweights q_0 toward R via the Gibbs tilt (2); the realized selection returns $\arg \max_i R(y_i)$ over the sampled pool, i.e. the $\beta \rightarrow 0^+$ oracle best-of- N . A deployable, *label-free* instance replaces the oracle reward by a minimum-Bayes-risk consensus score over $C(x)$, selecting the candidate of least expected disagreement with its peers.

Two properties of Definition 3 matter for the analysis. First, the reward truly *selects*: it picks among candidates the model itself sampled, so the operator realizes the tilt (2) rather than merely being scored by R . Second, its usefulness is entirely governed by how much the reward *varies across the sampled pool*: if the reward is nearly constant over $C(x)$ there is no headroom, whereas a reward that separates good from bad samples lets selection concentrate mass on the reward maximizer and the headroom grows with N . We make this dependence explicit in Section 2.3. The oracle uses the reference and is therefore an upper bound (not deployable); the consensus selector uses no labels and is the deployable proxy that realizes a fraction of that headroom.

Secondary operator: frozen-encoder probing (not Gibbs-tilt selection). The second operator is diagnostic, not generative, and we are explicit that it does *not* perform reward-driven selection.

Definition 4 (Frozen content-encoder probe). Let E be a frozen content bi-encoder (in our experiments omni-embed-nemotron-3b, a SentenceTransformer) mapping an utterance to an ℓ_2 -normalized embedding $E(x)$. We analyze E by *probing*: cosine classification against fixed label prototypes, and linear / k -NN probes trained on a held-out split for a target attribute (speaker identity, emotion, intent). The reported figure of merit is a verifiable *evaluation* accuracy against ground truth; it measures what the frozen embedding linearly exposes. Crucially, this accuracy is a *metric on a fixed readout*, not a reward that tilts a distribution over model-sampled candidates: there is no q_0 over generated outputs and no (2) tilt. Zero-shot classification via candidate-card cosine retrieval is a special case of this readout—a card-representation choice scored by the evaluation metric—and is likewise *not* training-free-RL selection.

Remark 1 (The two operators are distinct). Definition 3 and Definition 4 must not be conflated. Only Definition 3 samples candidates from a frozen *generative* model and lets a verifiable reward select among them, realizing the tilt (2); it is the paper’s genuine—though modest—training-free-RL result. Definition 4 is an analysis of a frozen *encoder*: its verifiable number is an evaluation metric, and any gain from a better card design (e.g. an altered query instruction or in-set example cards) is a zero-shot-classification effect that is confounded, not a reward-driven selection gain. We make no claim of “identical machinery” between the two, and we headline no probe number as an RL selection gain.

2.3 Reward spread

Best-of- N selection helps exactly to the extent that the reward is *not flat* across the model’s sampled candidate pool. We quantify this with the reward spread, the object that the qualitative lens in the Theory section ties to the optimization gain.

Definition 5 (Reward spread and optimization gain). Fix x , q_0 , R , and β . The *reward spread* spread is the range of R over the support of the prior, $\text{spread} = \max_{z \in \text{supp}(q_0)} R(z) - \min_{z \in \text{supp}(q_0)} R(z)$ (to leading order the q_0 -weighted variance $\text{Var}_{q_0}(R)$ plays the same role), and the *optimization gain* of training-free RL over the prior is

$$\mathcal{G} = F(q^*) - F(q_0) = \beta \log Z - \mathbb{E}_{z \sim q_0}[R(z)] \geq 0, \quad (3)$$

with equality if and only if R is q_0 -almost-surely constant.

Equation $(\star)$ is the free-energy expression for the value of the tilt: $\mathcal{G}$ is nonnegative, vanishes exactly when the reward carries no spread, and grows as the reward separates candidates. The Theory section develops this as a qualitative lens (the Donsker–Varadhan / free-energy identity, used to explain—not to newly prove—why gains appear) and pins its two endpoints with strict, machine-checked lemmas: a flat reward yields zero gain (`flat_no_gain`), and a non-constant reward yields strictly positive gain (`gain_pos_of_nonconstant`). This lens genuinely governs the primary result: the oracle best-of- N headroom on the frozen generative model is set by the spread of the verifiable reward across the model’s *sampled* candidates—flat reward across candidates means no headroom, while spread yields headroom that grows with N , exactly the empirical best-of- N shape we report. It also reads the encoder contrast qualitatively: the frozen content encoder exposes reward spread for content/intent attributes but little for paralinguistics, so probing recovers content/intent well and paralinguistics near chance.

2.4 Verifiable rewards for both operators

The validity of the guarantees in Sections 2.1 and 2.3 rests on R being trustworthy. We admit only *verifiable* rewards for both operators—negative word error rate as the selecting reward for the generative best-of- N (Definition 3), and probe/classification accuracy as the evaluation metric for the encoder (Definition 4).

Definition 6 (Verifiable reward). A reward R is *verifiable* if it is a deterministic, label-derived function of the output and a ground-truth annotation y^* , computable without invoking a learned scorer. Canonical instances are (i) negative word error rate $1 - \text{WER}(y, y^*)$ for ASR/ST, (ii) exact-match/accuracy for classification and intent, and (iii) retrieval hit@ k / probe accuracy against a fixed label. A reward is *model-judged* if it is the output of a separate generative or preference model evaluating y .

Assumption 1 (Verifiability). Throughout, R is verifiable in the sense of Definition 6: bounded in $[0, 1]$, deterministic given y^* , and independent of g_θ .

We adopt Assumption 1 for two reasons. First, model-judged rewards are susceptible to *reward hacking*—the optimizer exploits gaps between the proxy and the true objective, so that increasing $\mathbb{E}_{q^*}[R]$ no longer increases task quality [ska, 2022, gao, 2022]. Because the tilt (2) sharpens precisely along the direction of R , any reward misspecification is amplified rather than averaged out as $\beta \rightarrow 0$; this matters most for the generative selection operator, where the reward genuinely drives the choice. Second, learned judges carry systematic artifacts—position bias and self-preference—that contaminate any candidate ranking [pos, 2024, sel, 2024]; verifiable rewards are invariant to these. Restricting to label-derived R is therefore not a convenience but a precondition for the no-weight-update invariant of Definition 2 to translate into genuine, contamination-free activation of pretrained competence [tul, 2024]. Because the deployable consensus selector of Definition 3 uses *no* labels, the gap between its realized gain and the oracle (label-using) headroom is reported honestly as a realized-versus-headroom gap, not hidden. With Definitions 1 to 6 and Assumption 1 in place, the notation required by the Theory section is complete.

3 Related Work

Our contribution sits at the confluence of three literatures: the theory of inference-time reward optimization as exponential tilting, the empirical study of in-context learning (ICL) in audio language models, and the geometric characterization of what frozen speech encoders represent. We review each in turn, emphasizing not a catalogue of methods but a single structural fact that our experiments make concrete: training-free RL, realized here as *verifiable-reward selection* over a fixed candidate set $\mathcal{Z}$ drawn from one frozen speech model, is bounded by the *spread* of the reward across those candidates. Where the reward spreads—content and intent selection over candidate cards—selection buys real gains; where it is nearly flat—paralinguistic factors the frozen read-out has largely pooled away—selection has nothing to exploit. This spread lens organizes both our positive result (frozen bi-encoder card selection for spoken-language understanding) and our committed negative (little training-free paralinguistic gain over the same encoder).

3.1 Inference-Time / Training-Free RL and the Tilting View

The unifying abstraction behind training-free RL is that a reward-guided search distribution q is, in its optimal form, an exponential tilt of the base policy: the KL-regularized objective $F(q) = \mathbb{E}_q[R] - \beta \text{KL}(q \parallel q_0)$ is maximized by the Gibbs measure $q^*(z) \propto q_0(z) \exp(R(z)/\beta)$, with partition function Z [kor, 2022]. This identity recasts a spectrum of seemingly disparate inference-time procedures as approximations to the same tilt, and it is the abstraction under which our own method—selecting the best candidate card under a verifiable cosine reward—sits. Best-of- N (BoN) sampling is the most studied instance: bei [2024] establish that the KL divergence between the BoN distribution and q_0 grows as $\log N - (N-1)/N$, giving a closed-form exchange rate between compute and the distributional shift purchased. ver [2025] sharpen this with a soft, temperature-controlled BoN whose suboptimality decays as $O(1/N)$, while ami [2025] recast BoN through a smoothing lens, bounding its regret against the Gibbs optimum and exposing when additional samples cease to help. The crucial—and for our purposes load-bearing—observation across this line is that *every* tilt reweights existing candidates: the support of q is contained in the support of q_0 . No amount of N creates mass where q_0 assigns none, and, more sharply, no reweighting helps when the reward is constant across the candidate set. Minimum Bayes risk (MBR) decoding [ich, 2025], including its specialization to ASR [mbr, 2025], and controlled decoding [mud, 2023] inherit the same ceiling: they are consensus or value-guided rerankings, powerful when a correct hypothesis is already present with non-negligible probability and inert otherwise.

A more recent strand attempts to extract reward signal without any labels at all. TTRL [zuo, 2025] and TPO [li2, 2025] use self-consistency or majority vote as a surrogate reward at test time, and JitRL [jit, 2026] performs gradient-free credit assignment over a frozen policy. These methods are attractive precisely because they touch no weights, but they sharpen rather than dissolve the support constraint: a self-generated reward can only amplify modes the model already prefers, which is why majority-vote rewards can entrench confident errors. The over-optimization literature quantifies the resulting hazard. gao [2022] show that reward and true objective diverge as KL grows, following a predictable scaling law; kha [2025] characterize the optimal effective N^* (the Best-of-Poisson regime) beyond which tilting harms. In our notation these dynamics are governed by the reward spread spread and the realized identity gain $\mathcal{G}$: the achievable gain is $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$, which vanishes exactly when the reward is flat and grows with its spread. This is the qualitative content of the Donsker–Varadhan / free-energy identity we use throughout—not as a novel theorem, but as a lens—and it is the same quantity a verifiable-reward selector is implicitly maximizing over its candidate cards.

These dynamics intersect the ongoing RLVR boundary debate, which directly concerns whether inference-time or verifiable-reward optimization can expand a model’s capabilities or merely re-allocate them. rlv [2025] argue that RLVR predominantly sharpens the base distribution rather than adding skills, a position echoed by the finding that even spurious or random rewards can improve benchmark scores by exploiting format and prior structure [spu, 2025]. The countervailing evidence that RLVR can induce genuinely new skills [new, 2026] does not contradict the tilting view so much as locate its exception: new behavior emerges only when the candidate set $\mathcal{Z}$ carries enough reward spread—i.e., when a better hypothesis had non-zero base mass that a default single-shot read-out rarely surfaced. This reweighting-versus-expansion axis is exactly the one our spread lens sharpens, and it predicts, correctly for our data, where frozen selection helps (high-spread content/intent) and where it does not (near-flat paralinguistics).

3.2 In-Context Learning in Audio LLMs

A natural alternative to reward-guided selection is to move the base distribution with in-context demonstrations, and the text-domain ICL literature offers a sobering prior. min [2022] demonstrate that demonstrations are largely label-insensitive: corrupting ground-truth labels barely degrades performance, implying that demonstrations specify the task and output format rather than teaching the input–output mapping. ali [2026] reach a congruent conclusion for instruction-following demonstrations, which fix format but not accuracy. ICL is moreover scale-gated—its more sophisticated, override-the-prior behaviors emerge only in larger models [wei, 2023]—a property that carries into audio, where few-shot competence appears as an emergent phenomenon in larger audio LLMs [mim, 2025]. The function- and task-vector view [tod, 2023, tas, 2025] explains the mechanism: demonstrations compress to a latent task direction that conditions decoding without altering the underlying perceptual representation. In-context steering can thus re-route *how* content is verbalized, but it

does not re-open perceptual channels the encoder has already discarded.

This perceptual ceiling is starkly visible in audio. A growing body of evidence shows that audio LLMs frequently “read rather than listen”: [vox \[2026\]](#) document that models answer paralinguistic questions from textual priors and transcript content rather than the acoustic signal, and [lis \[2025\]](#) find that performance often depends on whether the model transcribes versus genuinely attends to non-lexical cues. In-context demonstrations do little to repair this when the requisite acoustic information has already been pooled away upstream—consistent with the broader finding that few-shot prompting fixes format, not the missing percept. SER is a partial and instructive exception: [ser \[2025\]](#) report that few-shot demonstrations *do* improve speech emotion recognition, plausibly because emotion is one of the paralinguistic factors an omni model retains with enough base probability to be surfaced. The pattern—ICL helping for retained factors, failing for discarded ones—mirrors the spread constraint of Section 3.1, and it foreshadows our committed negative: over a frozen content-oriented encoder, the paralinguistic reward has little spread for any training-free procedure, selection or ICL, to exploit.

The contrast with embedding models is relevant because our method uses precisely such a model—a frozen bi-encoder—as the selector. Text embedders do not acquire ICL for free: [bge \[2024\]](#) show that exploiting in-context examples in an embedder requires dedicated training, and instruction-conditioned embedders such as INSTRUCTOR [\[su2, 2022\]](#) must be explicitly trained to follow task instructions at encode time. An off-the-shelf embedding encoder therefore exposes no prompt-conditioned degree of freedom: its available action space is the choice of query, pooling, and layer, plus which candidate card to rank first—not a re-conditioning of the forward pass. Our selector operates entirely within this space, which is why the reward spread available in the encoder’s fixed geometry is the binding constraint on what training-free selection can deliver.

3.3 Embedding Geometry and Probing

Whether a frozen encoder *can* be tilted toward a task ultimately depends on whether the relevant information survives in its representations—a question the probing and embedding-geometry literatures answer with unusual clarity, and one that underpins our paralinguistic negative. Contrastive and alignment objectives are now understood to actively discard nuisance variation: [xia \[2020\]](#) show that contrastive learning suppresses factors not aligned with the training augmentations, and [wan \[2020\]](#) formalize the alignment–uniformity trade-off that drives semantically similar inputs together while spreading the hypersphere uniformly. For a content-oriented encoder, speaker identity, emotion, and channel are precisely the nuisances such objectives are designed to remove. This is the geometric root of why a frozen selector, however cleverly searched, cannot recover a factor its training objective erased: the base distribution q_0 over the candidate set simply carries no reward spread along the discarded direction.

Pooling compounds the loss. Mean- or last-token pooling imposes an information bottleneck that NV-Embed’s learned latent-attention pooling was introduced to mitigate [\[nve, 2024\]](#), and the speaker-verification community has long known that attentive statistics pooling [\[att, 2018\]](#) and ECAPA-TDNN’s multi-scale aggregation [\[eca, 2020\]](#) are necessary precisely because naive temporal averaging destroys speaker-discriminative structure. The implication is that the *same* encoder can be near-blind or highly informative for a factor depending solely on the read-out. Layer-wise probing makes this concrete: [pas \[2021\]](#) chart an autoencoder-like trajectory across wav2vec 2.0 layers in which phonetic and word-level information peaks at intermediate depths, and [spk \[2025\]](#) show speaker attributes are distributed non-uniformly across layers. Capability is thus not a scalar property of a model but a function of (layer, pooling, query). Our committed probe of the frozen omni content read-out is consistent with this picture from the pessimistic side: on CREMA-D it recovers speaker identity only near chance and emotion only modestly, so a training-free selector operating on that read-out inherits a near-flat paralinguistic reward and, empirically, no significant emotion gain.

Evidence that paralinguistic content *persists* elsewhere in a model, even when unused, is nonetheless substantial. [fro \[2024\]](#) demonstrate that a frozen LLM, given speech features, perceives paralinguistic attributes it never verbalizes; [emo \[2026\]](#) localize emotion-selective units inside such models. We read these results carefully: they show the information can survive in *some* read-out, not that it is available in the particular content-oriented encoder our selector uses, and our negative quantifies exactly that gap for one

committed configuration. The SUPERB and Dynamic-SUPERB taxonomies [sup, 2021, dyn, 2023] supply the benchmark scaffolding for evaluating capability across content, speaker, semantic, and paralinguistic axes, and we adopt their factorization to structure our measurements. What this literature lacks—and what our spread lens supplies—is a principled account of *which* latent capabilities are reachable by reweighting a fixed candidate set and which demand a richer read-out or a different action space altogether.

3.4 Cross-Session Agent Memory and Skills (Future-Work Context)

Our contributions are confined to a single frozen speech model and inference-time verifiable-reward *selection*; we make no agent-level claim. For completeness we note the literature that would motivate a *deferred*, out-of-scope program on cross-session speech agents, and we are explicit that none of it is related to a result we report here. A line of work on training-free agent memory accumulates experience without touching weights — from raw interaction streams and paged context [gen, 2023, mem, 2023] to first-class curation operators [mem, 2025a], with reinforcement learning used to assign credit over memory-management actions [mem, 2025b]; a parallel line accumulates reusable, reward-verified skills atop frozen models [voy, 2023, sok, 2026]. Evaluation of such systems remains almost entirely text-centric and single-episode: long-horizon and conversational probes exist [lon, 2024, loc, 2024], audits show many ostensibly memory-dependent tasks need no genuine cross-session recall [ana, 2026], and spoken benchmarks evaluate a fresh agent per interaction [uro, 2025, voi, 2024] with no isolated cross-session, same-speaker suite [amb, 2026].

We flag this only to scope our reward-spread lens honestly. Extending selection to an agent that persists across sessions would require three ingredients absent from our setting: (a) a cross-session, same-speaker corpus (none exists in our frozen data); (b) a locally-driven generative operator (we run only a frozen bi-encoder over candidate cards, never a local generative model); and (c) a genuinely new, non-separable irreducibility result. Absent (c), the spread lens implies context isolation adds nothing on a fixed reward: a memory that merely re-partitions the candidate pool cannot raise the identity gain beyond what the reward spread already permits. We therefore leave the agent-level program to future work and claim none of it here.

4 A Reward-Spread Lens

We keep one lightweight analytic tool, used *qualitatively* to organize our empirical results — not as a novel theorem, and not as a “machine-checked quantitative ceiling.” The claim is simple: inference-time verifiable-reward selection over a frozen model can only realize gain to the extent that the reward actually *spreads* across the candidates it ranks. It applies directly to the genuine training-free RL of Section 5.2: the oracle best-of- N headroom on a frozen generative omni model is governed by how much the verifiable reward varies across the transcripts the model itself samples. Where the reward has spread, best-of- N has headroom that grows with N ; where it is nearly flat, there is nothing for selection to exploit.

Fix a finite candidate set $\mathcal{Z}$ (for Section 5.2, the N transcripts a frozen generative model samples for one utterance), a strictly positive base measure $q_0 : \mathcal{Z} \rightarrow \mathbb{R}_{>0}$ with $\sum_{w \in \mathcal{Z}} q_0(w) = 1$ (the model’s own sampling distribution over candidates), a bounded verifiable reward $R : \mathcal{Z} \rightarrow \mathbb{R}$ (for ASR, negative per-candidate WER), and an inverse-temperature $\beta > 0$. Write $\mathbb{E}_q[\cdot]$ for expectation under q , adopt $0 \log 0 = 0$, and define the tilted (Gibbs) target and its normalizer

$$Z := \mathbb{E}_{q_0}[\exp(R/\beta)] > 0, \quad q^*(z) := \frac{q_0(z) \exp(R(z)/\beta)}{Z},$$

together with the KL-regularized objective $F(q) := \mathbb{E}_q[R] - \beta \text{KL}(q \| q_0)$. This is the standard inference-time specialization of KL-regularized reward maximization, whose maximizer is the exponential tilt [kor, 2022, mud, 2023]; the reward-argmax best-of- N we run in Section 5.2 is its $\beta \rightarrow 0$ limit.

Master identity: gain is a KL divergence. A one-line algebraic identity (substitute $R(z) = \beta(\log \frac{q^*(z)}{q_0(z)} + \log Z)$ into $F(\cdot)$) gives, for every distribution q on $\mathcal{Z}$,

$$F(q) = \beta \log Z - \beta \text{KL}(q \| q^*).$$

Reading it at the model’s own distribution $q = q_0$ and defining the optimization *gain* $\mathcal{G} := F(q^*) - F(q_0)$ yields the object we actually care about:

$$\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*) = \beta \log \mathbb{E}_{q_0}[\exp(R/\beta)] - \mathbb{E}_{q_0}[R]. \quad (\star)$$

The gain is exactly the divergence between what the model already does, q_0 , and what the reward would have it do, q^* . The right-hand form of $(\star)$ is the free-energy / cumulant expression $\beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R]$; by the Donsker–Varadhan variational identity it is nonnegative, and it is governed by how much R varies under q_0 . We use this *qualitatively*: a second-order expansion gives $\mathcal{G} \approx \text{Var}_{q_0}(R)/(2\beta)$ for a concentrated q_0 , so the gain scales with the *spread* spread of the reward across candidates. This is a well-known identity, invoked here as an organizing lens, not billed as a new result or a tight ceiling.

Two endpoints (both sorry-free in Lean). The lens has two exact endpoints, which we have formalized in Lean 4 without `sorry`.

(i) *Flat reward $\Rightarrow$ zero gain* (`flat_no_gain`). If $R \equiv c$ is constant on $\mathcal{Z}$, then $Z = e^{c/\beta}$, so $q^* = q_0$ and, by $(\star)$, $\mathcal{G} = 0$. A reward that cannot distinguish the candidates leaves the model unchanged: there is nothing for selection to exploit — when every sampled transcript earns the same WER, best-of- N returns the greedy answer.

(ii) *Non-constant reward $\Rightarrow$ strictly positive gain* (`gain_pos_of_nonconstant`). If R is non-constant on $\mathcal{Z}$ (recall $q_0 > 0$ everywhere), then $q_0 \neq q^*$: were they equal, $e^{R(z)/\beta} = Z$ for all z would force R constant. By the strict Gibbs inequality $\text{KL}(q_0 \parallel q^*) > 0$, so $(\star)$ gives $\mathcal{G} > 0$. Any genuine spread in the verifiable reward yields strictly positive headroom for selection to realize.

These two lemmas are the strict converse of one another: $R \text{ constant} \iff q^* = q_0 \iff \mathcal{G} = 0$.

Why the lens explains our results. The lens reads directly onto the genuine best-of- N of Section 5.2. On the frozen generative omni model, the oracle-WER selector (the $\beta \rightarrow 0$ tilt) reduces WER from a greedy 0.117 to 0.073, a significant +0.044 reduction (95% CI [0.024, 0.067]), and this headroom *grows with* N : +0.013, +0.032, +0.044 at $N = 2, 4, 8$ (Table 1). Endpoint (ii) is exactly this shape — the model samples transcripts on which the verifiable reward has real spread, so there is strictly positive headroom, and drawing more candidates ($N \uparrow$) exposes more of that spread and enlarges the gain, the inference-time analogue of why best-of- N and reranking pay off in proportion to how much a verifiable reward separates candidates [bei, 2024, gao, 2022]. Endpoint (i) marks the ceiling: were every sampled transcript equally (in)correct, the reward would be flat and no selector — oracle or deployable — could beat greedy. The deployable label-free MBR-consensus selector realizes only a small, boundary-significant fraction of this headroom (WER 0.102, +0.014, 95% CI [−0.0001, 0.033]), an honest realized-vs-headroom gap: the reward spread is present, but a reference-free proxy recovers little of it.

The same lens organizes the frozen-encoder probing of Section 5.3 qualitatively. The frozen content bi-encoder exposes reward spread for *content and intent* — its embedding geometry separates transcript- and intent-bearing candidates sharply — but carries near-chance speaker and only modest emotion information (Table 2), so on paralinguistic factors the corresponding reward is nearly flat on $\text{supp}(q_0)$ and training-free pool-selection produces no significant emotion gain (across-seed 95% CI [−0.043, +0.116]; committed artifact `_repro/emotion_pool_paired_v2.json`). By endpoint (i) there is little spread to exploit and selection cannot help — exactly what the probe observes. The reward-spread lens thus does one job, and only that job: it explains why verifiable-reward selection has genuine, N -scaling headroom on ASR content and why a frozen content encoder surfaces content/intent but not paralinguistics, without appeal to model class or agentic structure.

5 Empirical Results

We report two results on frozen omni speech *models*, each obtained on a single frozen model with *no weight updates and no structural change*. The primary result is genuine training-free RL: on a frozen *generative* omni model we sample N candidate transcripts per utterance and select among them with a verifiable

reward—reward-driven best-of- N over model-sampled candidates, on a frozen model (Section 5.2). The secondary result is honest *probing* of a different frozen model, a content bi-encoder: a paralinguistic read-out and a null emotion-selection gain, with an accompanying zero-shot-classification ablation (Section 5.3). The two are read through a single quantitative lens—the *spread* of the verifiable reward across candidates—which we state first (Section 5.1). We are careful to keep the operators distinct: the primary result is reward-driven selection over *generated* candidates; the encoder probe is not reward-driven selection at all, and we make no claim of shared machinery between them.

5.1 The Reward-Spread Lens

The optimization performed by reward-driven best-of- N is the KL-regularised tilt of a frozen base distribution q_0 toward a verifiable reward R : maximising $F(q) = \mathbb{E}_q[R] - \beta \text{KL}(q \| q_0)$ has optimum $q^* \propto q_0 \exp(R/\beta)$, the object that underwrites best-of- N and reward-guided decoding [kor, 2022, mud, 2023, bei, 2024]. As $\beta \rightarrow 0$ this tilt concentrates on the highest-reward candidate—the argmax that oracle best-of- N realises. The gain of the tilt admits the identity

$$\mathcal{G} = \beta \text{KL}(q^* \| q_0),$$

which we use *qualitatively*: it is the Donsker–Varadhan / free-energy identity for the log-partition function $\log Z = \log \mathbb{E}_{q_0}[\exp(R/\beta)]$, not a novel theorem of ours. Its content for us is a comparison, not a constant: the gain is controlled by the reward *spread* $\text{spread} = \max R - \min R$ over the candidates in $\text{supp}(q_0)$. A flat reward—one that does not discriminate among candidates—yields *zero* gain; a reward with genuine spread across the model’s sampled candidates yields *strictly positive* gain, and does so on a single frozen model, with no decomposition or additional structure required. Two strict, *sorry-free* Lean lemmas make exactly these two endpoints precise: `flat_no_gain` (a constant reward gives $\mathcal{G} = 0$) and `gain_pos_of_nonconstant` (a non-constant reward gives $\mathcal{G} > 0$). We claim nothing beyond these two facts; they suffice to organise the evidence.

The lens speaks directly to the primary result: the oracle best-of- N headroom is governed by the spread of the verifiable reward across the model’s sampled candidates. When the candidates agree (little spread), there is no headroom; when they disagree in a way the reward can score (spread), there is headroom, and it grows as N enlarges the candidate set and thus the achievable spread. The same lens frames the secondary contrast qualitatively: the frozen content encoder exposes reward spread for content and intent, but little for paralinguistics, so a probe over it finds structure in the former and near-none in the latter.

5.2 Genuine Reward-Guided Best-of- N on a Frozen Omni Model

Our primary result is genuine training-free RL: reward-driven selection over candidates that a frozen generative model actually samples. On **Qwen3-Omni-30B-A3B** (Q8_0 GGUF, driven by `llama.cpp`) we sample N candidate transcripts per utterance and select one under a verifiable reward—no weights are touched and no structure is added. We evaluate on LibriSpeech `test-other` with additive noise at SNR=5 dB, a robustness perturbation chosen because it leaves measurable headroom over greedy decoding; $n=96$ utterances, sampling pool 8, temperature 0.8, seed 42. Confidence intervals are paired bootstraps on the per-utterance WER reduction against greedy (committed artifact `_repro/asr_bon_llamacpp_snr5.json`).

We report two selectors. The *oracle-WER* selector takes the pool candidate of minimum WER against the reference: it is the verifiable-reward argmax—the $\beta \rightarrow 0$ tilt—and it is an *upper bound*, not deployable, because it consults the reference. The *MBR-consensus* selector is reference-free and therefore deployable: it selects the candidate of minimum expected risk against the rest of the pool, using no labels. Table 1 reports both across N .

The verifiable-reward headroom is real, significant, and N -scaling on a frozen model. Greedy WER is 0.117; oracle best-of- N reduces it to 0.073 at $N=8$, a reduction of +0.044 (95% CI [0.024, 0.067], significant), and the headroom grows monotonically with N (+0.013, +0.032, +0.044 at $N=2, 4, 8$; all significant from $N=2$). This is exactly the empirical best-of- N shape the lens predicts: enlarging the sampled pool enlarges the reward spread the argmax can exploit. This is genuine training-free RL—reward-driven selection over model-sampled candidates, with no weight updates.

Table 1: Reward-guided best-of- N on a frozen Qwen3-Omni-30B-A3B (Q8_0 GGUF via llama.cpp), LibriSpeech `test-other+noise` (SNR 5dB), $n=96$, pool 8, temp 0.8, seed 42; greedy baseline WER 0.117. Reductions are vs. greedy with paired-bootstrap 95% CIs (committed artifact `_repro/asr_bon_llamacpp_snr5.json`). The oracle-WER selector is the verifiable-reward argmax (an upper bound, uses the reference); MBR-consensus is the deployable, label-free proxy. The oracle headroom is significant from $N=2$ and scales with N ; the deployable MBR selector is boundary non-significant.

N	Oracle-WER (verifiable-reward argmax)			MBR-consensus (deployable, label-free)		
	WER	Δ (reduction)	95% CI, sig.	WER	Δ (reduction)	95% CI, sig.
1	0.109	+0.007	$[-0.003, 0.021]$ n.s.	0.109	+0.007	$[-0.003, 0.021]$ n.s.
2	0.103	+0.013	$[0.002, 0.028]$ SIG	0.108	+0.009	$[-0.002, 0.022]$ n.s.
4	0.085	+0.032	$[0.014, 0.053]$ SIG	0.107	+0.010	$[-0.001, 0.024]$ n.s.
8	0.073	+0.044	$[0.024, 0.067]$ SIG	0.102	+0.014	$[-0.0001, 0.033]$ n.s.

Table 2: Frozen content read-out on CREMA-D (linear/kNN probe, dev-selected layer, 3 seeds; committed artifact `_repro/paralinguistic_negative_probe.json`). The content embedding carries near-chance speaker and only modest emotion information: little paralinguistic spread for a selector to exploit.

Factor	Ways	Chance	Mean acc	$\times$ chance
Speaker	91	0.011	0.033	$3.0\times$
Emotion	6	0.167	0.405	$2.4\times$

We are equally plain about the realized-vs-headroom gap. The deployable, label-free MBR-consensus selector reaches WER 0.102 at $N=8$, a reduction of +0.014 (95% CI $[-0.0001, 0.033]$, boundary non-significant). It realises only a small fraction of the oracle headroom: the label-free consensus proxy is a much weaker stand-in for the verifiable reward than the reference-based argmax. The honest reading is that the reward-defined headroom is significant and grows with N , while a deployable selector that never sees labels captures only a boundary-significant sliver of it.

Three caveats we state rather than bury. (i) llama.cpp audio input is *experimental* upstream and documented as possibly reduced quality; the absolute WERs should be read with that in mind, though the *within-setup* best-of- N contrast is the load-bearing quantity. (ii) The SNR-5dB additive noise is a deliberate robustness perturbation chosen to expose headroom, not a clean-speech benchmark. (iii) The oracle selector uses the reference and is therefore an upper bound, not a deployable system; MBR is the deployable proxy, and the gap between them is precisely what the table quantifies. Reproduce with the committed command `scripts/repro_asr_best_of_n_llamacpp.py` (with the llama-server resident) against the committed artifact.

5.3 Frozen-Encoder Probing: A Paralinguistic Negative

Our secondary result probes a *different* frozen model and a *different* operator. On a frozen content bi-encoder (omni-embed-nemotron-3b) we analyse what its embeddings carry by probing—not by reward-driven selection. This is not the best-of- N operator of Section 5.2, and we do not claim shared machinery; it is a measurement of the frozen representation.

First, the frozen content read-out carries little paralinguistic information. On CREMA-D, with the read-out layer selected on dev and three seeds, speaker identity is recoverable at *near chance*—mean accuracy 0.033 on a 91-way task ($3.0\times$ chance, chance 0.011)—and emotion only modestly—mean accuracy 0.405 on a 6-way task ($2.4\times$ chance, chance 0.167) (committed artifact `_repro/paralinguistic_negative_probe.json`). Table 2 summarises. For a *content* embedding, near-chance speaker recovery is close to definitional; an index built on this embedding would retrieve on content, not on who-spoke or how-they-felt.

Second, training-free selection over this encoder yields *no significant* paralinguistic gain. Treating the choice of read-out (layer and pooling) as the selectable degree of freedom, and evaluating the emotion

Table 3: CREMA-D emotion, training-free pool-selection (mean-pool baseline vs. attentive-statistics pooling, both layers dev-selected, paired-delta bootstrap on TEST, five seeds; committed artifact `_repro/emotion_pool_paired_v2.json`). The across-seed 95% t -CI $[-0.043, +0.116]$ spans zero: no significant emotion gain. The earlier single-seed $+0.097$ was the best seed under oracle test-layer selection, reproduced here only as the best of five.

Seed	base (dev-sel)	sel (dev-sel)	acc _{base}	acc _{sel}	Δ	paired 95% CI
42	audio-mean L0	audio_attn L16	0.400	0.497	$+0.097$	$[+0.027, +0.167]$
7	audio-mean L0	audio_attn L0	0.423	0.370	-0.053	$[-0.117, +0.010]$
123	audio-mean L0	audio_attn L16	0.390	0.473	$+0.083$	$[+0.013, +0.150]$
2024	audio-mean L0	audio_attn L16	0.493	0.487	-0.007	$[-0.077, +0.063]$
31337	audio-mean L0	audio_attn L16	0.400	0.463	$+0.063$	$[-0.010, +0.130]$
mean	—	—	—	—	$+0.037$	across-seed 95% t -CI $[-0.043, +0.116]$ (spans 0; nu

Table 4: Zero-shot SLU intent *classification* on MInDS-14 en-US tool/intent by cosine retrieval against candidate cards (frozen `omni-embed-nemotron-3b`, $n=182$, seed 42; committed artifact `_repro/minds14_toolintent_paired.json`). This is a card-representation, zero-shot-classification ablation—the verifiable reward is the evaluation metric only and does *not* drive candidate selection. The policy-card gain is confounded (in-set example cards + a query-instruction change) and is *not* a training-free-RL selection gain; it is a distinct operator from the generative best-of- N of Section 5.2.

Task (n)	Candidate-card representation	Acc@1	Δ vs. baseline	95% CI
MInDS-14 (182)	naive card	0.720	—	—
	raw-schema (baseline)	0.857	—	—
	policy (boundary-aware, <i>confounded</i>)	0.984	$+0.126$	$[0.077, 0.181]$

improvement of attentive-statistics pooling over a mean-pool baseline—both layers selected on dev and frozen before test, with a paired-delta bootstrap and five seeds {42, 7, 123, 2024, 31337}—the across-seed mean improvement is $\Delta = +0.037$ with a 95% t -CI of $[-0.043, +0.116]$, which *spans zero* (committed artifact `_repro/emotion_pool_paired_v2.json`). Table 3 reports the per-seed detail. The headline is a null: training-free pool-selection produces no significant emotion gain. We note plainly that an earlier single-seed report of $+0.097$ was the best seed under *oracle test-layer* selection (a selection leak); under dev-selection it reproduces only as the best of five seeds, and the appropriate across-seed aggregate is the null above.

An honest note on the same encoder, which we deliberately do *not* headline as a selection or RL gain. The frozen bi-encoder also does zero-shot SLU intent classification by cosine retrieval against candidate *cards* (textual schemas describing each intent): on MInDS-14 en-US tool/intent it reaches $\text{Acc@1} = 0.984$ ($n=182$, seed 42; committed artifact `_repro/minds14_toolintent_paired.json`). We frame this precisely for what it is. It is a *card-representation, zero-shot-classification* result: the verifiable reward here is only the evaluation metric—it scores the prediction, it does *not* drive a selection over generated candidates—so it is a distinct operator from the generative best-of- N of Section 5.2, and we claim no shared machinery. Table 4 reports the card-design ablation under the fixed cosine rule; we read it as a zero-shot-classification ablation, not a training-free-RL selection gain. The apparent card-design gain is moreover *confounded*: the “policy” card both injects in-set example cards and changes the query instruction, so the $+0.126$ over the raw-schema card mixes representation and prompt effects and is not a clean attribution. We report it plainly and draw no reward-selection conclusion from it. (We drop an earlier SLURP intent number, which has no committed artifact.)

We keep the scope of the paralinguistic negative tight. It is a linear/kNN probe and a pool-selection experiment on *one* frozen content bi-encoder; it does not generalise to frozen omni models as a class, and CREMA-D content is a 12-sentence closed read set, so we make no content claim from it. What it establishes is precisely what the spread lens predicts: this content encoder presents little paralinguistic spread, so probing over it finds near-none.

5.4 Reading the Evidence

The two results are read through one lens, but they are two different operators on two different frozen models, and we do not conflate them. The primary result is genuine training-free RL: reward-driven best-of- N over candidates a frozen generative omni model actually samples, where the verifiable-reward headroom is real, significant, and N -scaling (+0.044, 95% CI [0.024, 0.067] at $N=8$), while the deployable label-free selector realises only a small, boundary-significant fraction of it (+0.014, 95% CI [-0.0001, 0.033])—an honest realized-vs-headroom gap. The secondary result is honest probing of a frozen content encoder: near-chance speaker recovery, only modest emotion, and a null emotion-selection CI [-0.043, +0.116]; the encoder’s strong zero-shot intent *classification* (0.984) is a card-representation result with a confounded gain, not a reward-driven selection gain. The gain identity $\mathcal{G} = \beta \text{KL}(\|q_0\| \|q^*\|)$ and its two Lean endpoints (`flat_no_gain`, `gain_pos_of_nonconstant`) organise both qualitatively: the best-of- N headroom tracks the spread of the verifiable reward across the model’s sampled candidates (spread grows with N , and so does the headroom), and the frozen content encoder exposes such spread for content and intent but little for paralinguistics. We report every interval in full and state the null as plainly as the positive.

6 Threats to Validity and Limitations

We hold this work to the standard of self-criticism it advocates, and enumerate the threats in descending order of how much they could change the paper’s claims. The scope is deliberately narrow: two *frozen omni speech models*, each carrying exactly one of our two results, with no weight update and no structural change. Our primary result is genuine training-free RL: on a *frozen generative omni model* (Qwen3-Omni-30B-A3B, Q8_0 GGUF driven by `llama.cpp` [qwe, 2025b]) we sample N candidate transcripts per utterance and *select* with a verifiable reward, i.e. best-of- N over model-sampled candidates. Our secondary result is *probing*, not RL: a frozen content bi-encoder (`omni-embed-nemotron-3b` [omn, 2025]) is analysed for what paralinguistic information its read-out affords. These are *different frozen models* and *different operators* — a generative best-of- N selector versus a static embedding probe — and we are careful throughout not to conflate them.

llama.cpp audio input is experimental upstream. The primary result runs the frozen omni model through `llama.cpp`, whose multimodal audio path is flagged experimental upstream and documented as possibly reduced quality relative to the reference runtime. Our greedy baseline WER of 0.117 on LibriSpeech test-other with additive noise is therefore a property of *this* decoding path, and the absolute error rates — though not necessarily the *relative* best-of- N headroom, which is a within-runtime comparison — may shift under a higher-fidelity runtime. Every number is committed to `_repro/asr_bon_llamacpp_snr5.json` with its reproduce command, so the runtime dependence is inspectable rather than hidden.

The noise condition is a chosen robustness perturbation; headroom may differ clean or on other corpora. We evaluate on LibriSpeech test-other with additive noise at SNR 5 dB, deliberately chosen to expose measurable best-of- N headroom: on clean, near-ceiling audio the greedy transcript is often already correct and the reward has little spread across sampled candidates to exploit. This means our headroom figures are conditional on a perturbation we selected *because* it creates spread. On clean audio, on other corpora, or under other noise types the oracle headroom and the realised deployable gain may both be smaller. We report the condition plainly rather than presenting it as a clean-speech result.

The deployable selector realises only a small, boundary-significant fraction of the oracle headroom. The honest core of the primary result is a *realised-versus-headroom gap*. The verifiable-reward argmax — oracle-WER best-of- N , the $\beta \rightarrow 0$ tilt of the reward-spread lens — reduces WER from 0.117 to 0.073 at $N=8$, a reduction of +0.044 (95% paired-bootstrap CI [0.024, 0.067], significant; $n=96$, pool 8, temperature 0.8, seed 42), and the headroom *scales* with N (+0.013, +0.032, +0.044 for $N=2, 4, 8$) — the empirical best-of- N shape. But this oracle uses the reference transcript to pick the best candidate, so it is an *upper bound*, not a deployable procedure. The deployable, label-free selector is MBR consensus, which uses no references and reaches WER 0.102, a reduction of only +0.014 (95% CI [-0.0001, 0.033]),

boundary non-significant at $n=96$). The correct reading is thus twofold: the verifiable-reward headroom is real, significant, and N -scaling on a frozen model (genuine training-free RL), while the label-free selector currently captures only a small, boundary-significant slice of it. We do not paper over that gap; closing it is the main deployable open problem (Section 7).

Possible pretraining contamination of the public sets. LibriSpeech, and any public set we touch, may overlap the pretraining corpora of the frozen models [con, 2024, gor, tas], so absolute error rates can in part reflect memorised material. Our primary claim is a *within-utterance, within-runtime* best-of- N contrast — how much a verifiable reward improves over the same model’s greedy output on the same items — which is more robust to contamination than an absolute score, and inference-time selection against a verifiable reward is by construction an over-optimization engine that will exploit any such regularity [gao, 2022, ska, 2022, spu, 2025]. A contamination-controlled split would still be needed to fully exclude leakage; until then the numbers should be read as within-model, within-condition contrasts.

The C2 negative is specific to *this* frozen content encoder, not to frozen models as a class. The secondary (probing) negative is precise and committed but narrowly scoped. On CREMA-D, a probe of the frozen `omni-embed-nemotron-3b content` read-out (dev-selected layer, three seeds, `_repro/paralinguistic_negative_probe.json`) sits near chance for speaker (mean accuracy 0.033, $3.0\times$ the 91-way chance rate) and is only modest for emotion (0.405, $2.4\times$ the 6-way chance rate); and training-free pool-selection over that read-out yields *no* significant emotion gain (across-seed 95% CI $[-0.043, +0.116]$, spanning zero, `_repro/emotion_pool_paired_v2.json`). This bounds *that content read-out* and nothing more. A purpose-built speaker encoder such as an ECAPA-TDNN embedding [eca, 2020] would carry speaker identity well — a well-established property of dedicated speaker models that we *cite* rather than claim to have run. The finding is consistent with evidence that omni models tend to “read, not listen” on paralinguistic cues [vox, 2026, lis, 2025] and that speaker identity resists in-context correction [spk, 2026].

The MInDS card comparison is a zero-shot-classification ablation, not a reward-driven selection gain. The same frozen content encoder also does zero-shot SLU/intent classification via candidate-card cosine retrieval, reaching MInDS-14 en-US Acc@1 0.984 (`_repro/minds14_toolintent_paired.json`). We report this plainly and only as a *card-representation / zero-shot-classification* result: its “verifiable reward” is merely the *evaluation* metric and does *not* drive any selection, so it is not best-of- N , not Gibbs tilting, and not reward-driven RL. It is also a *distinct operator* from the C1 generative best-of- N — we make no claim of identical machinery between them. Moreover the apparent card-design improvement is confounded: the “policy” card differs from the baseline both by including in-set example cards and by a query-instruction change, so any Acc@1 delta cannot be cleanly attributed to a selection mechanism. We therefore do not headline it as an RL/selection gain, and we drop the earlier SLURP intent figure entirely, as it has no committed artifact.

Relation to best-of- N / MBR / reranking (honest differentiation). The ingredients of the primary result — best-of- N and reranking [bei, 2024, ver, 2025, gui, 2024], MBR decoding and MBR-for-ASR [jin, 2024, ich, 2025, mbr, 2025], controlled decoding [mud, 2023], test-time scaling [sne, 2024], and the Gibbs-optimal tilt $q^* \propto q_0 \exp(R/\beta)$ from alignment-as-inference [kor, 2022, yan, 2024] — are all public, and we claim none as novel. What is specific here is applying a genuine verifiable-reward best-of- N to a *frozen omni generative model* and reporting the honest realised-versus-headroom gap between the oracle upper bound and a label-free deployable selector, alongside a committed paralinguistic *probing* negative on a second frozen model and the reward-spread lens that organises both. The contribution is integrative and empirical, and should be weighted as such.

Goodhart pressure on the oracle reward. Pushing N moves probability mass toward whatever the reward measures, including its defects [gao, 2022, ska, 2022, rlv, 2025]. Our oracle reward is WER against the reference: verifiable and hard to game, but definitionally an upper bound that a deployed system cannot compute. The deployable MBR proxy is label-free precisely to avoid this, at the cost of the boundary

significance reported above. Because we change no weights, the failure mode is recoverable (lower N , better proxies); the lens simply predicts that no selector can exceed the oracle headroom, which is itself bounded by the spread of the reward across the model’s sampled candidates.

Future work. Two directions follow directly and honestly. First, *stronger deployable selectors* — learned rerankers or reference-free quality estimators trained to approximate the verifiable reward — to close the realised-versus-headroom gap left open by MBR consensus. Second, *larger and clean-condition best-of- N* — more candidates, clean audio, and additional corpora — to map how the oracle headroom varies with condition and N . A larger agent-level program (one frozen model routing and rewarding another across sessions) remains deferred, not disproved: it would require a cross-session, same-speaker corpus we do not have, and a genuinely new non-separable irreducibility theorem showing that decomposing a task into isolated contexts can add optimization headroom a single model lacks. Absent such a theorem, the spread lens is explicit that context isolation adds *nothing* on a fixed reward — no new non-constant reward component, no new spread, and by the identity below no new gain — so that theorem is the precondition for any agent-level claim, and it is exactly what our present data do not supply.

7 Conclusion

We set out to test whether the gain available to training-free RL — inference-time selection under a verifiable reward, changing no weights and no structure — is real on frozen omni speech models, and whether it is governed by the *spread* of that reward across candidates. Our primary result answers the first question affirmatively and modestly. On a frozen generative omni model (Qwen3-Omni-30B-A3B via `llama.cpp`) we sample N transcripts per utterance on LibriSpeech test-other with additive noise and select with a verifiable reward — genuine best-of- N over model-sampled candidates. The verifiable-reward argmax reduces WER from 0.117 (greedy) to 0.073 at $N=8$, a significant +0.044 reduction (95% CI [0.024, 0.067]) whose headroom scales with N (+0.013, +0.032, +0.044 for $N=2, 4, 8$; committed artifact `_repro/asr_bon_llamacpp_snr5.json`). This is genuine training-free RL: reward-driven selection over a frozen model’s own samples, no weights touched. The label-free, deployable MBR-consensus selector realises only a small fraction of that headroom, reaching WER 0.102 (+0.014, 95% CI [−0.0001, 0.033], boundary non-significant at $n=96$) — an honest realised-versus-headroom gap rather than a solved deployment.

Our secondary result is *probing*, on a *different* frozen model and via a *different* operator: a frozen content bi-encoder (`omni-embed-nemotron-3b`) is near chance for speaker (mean 0.033, $3.0\times$ the 91-way chance rate) and only modest for emotion (0.405, $2.4\times$ the 6-way chance rate) on CREMA-D, and training-free pool-selection over its content read-out yields no significant emotion gain (across-seed CI [−0.043, +0.116]). The same encoder does high-accuracy zero-shot intent classification by candidate-card cosine retrieval (MInDS-14 Acc@1 0.984), but that is a card-representation / zero-shot-classification observation whose reward is only the evaluation metric and whose card-design delta is confounded — *not* a reward-driven selection gain, and not the same machinery as the generative best-of- N above. We report the paralinguistic negative not as a setback but as a precise, committed measurement that scopes what this frozen *content* read-out affords.

The lens that ties these together is deliberately lightweight and used qualitatively. The identity $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$ — the free-energy / Donsker–Varadhan characterisation of the gap between what the frozen model already does (q_0) and what the reward would have it do (q^*) — is a known result, not a novel theorem, and we invoke it to *explain*, not to prove: the gain is bounded by the reward spread over $\text{supp}(q_0)$, so a flat reward yields vanishing gain while a high-spread reward yields large gain, on the very same frozen model. Two strict, *sorry-free* Lean lemmas anchor its endpoints: a flat reward forces zero gain (`flat_no_gain`), and a non-constant reward forces strictly positive gain (`gain_pos_of_nonconstant`). This is precisely the contrast we observe. In the primary result the oracle best-of- N headroom is governed by the spread of the verifiable reward across the model’s sampled candidates — flat reward across candidates would give no headroom, and the observed spread gives headroom that grows with N . In the secondary result the frozen content encoder exposes reward spread for content and intent but little for paralinguistics, so selection over it has purchase in the former and essentially none in the latter.

The conclusion is therefore modest and, we believe, true. A verifiable-reward best-of- N on a frozen omni

model yields significant, N -scaling headroom — genuine training-free RL; a label-free deployable selector currently realises only a small fraction of it; a frozen content encoder is near chance on paralinguistics; and the reward-spread lens explains both. What we offer is a small, reproducible set of results and a lens that predicts their sign — not a deployable agentic stack, and not a claim beyond the frozen models and committed artifacts we can currently stand behind.

A Lean 4 Formalization

The lens of this paper — that the optimization gain of training-free RL (inference-time verifiable-reward selection) is governed by the *reward spread*, not by any model class — rests on one qualitative identity, and *that* identity is machine-checked in **Lean 4 + mathlib** (toolchain and mathlib both pinned to **v4.31.0**; the exact mathlib commit is recorded in the `lake-manifest.json` committed alongside the proof project, see Section A), in the project `proofs/tf1` (Lean namespace `Tf1Proofs`). We are careful about what “machine-checked” means, and Section A below states it precisely. The retained core is small: the master identity and T1 optimality, the flat-reward collapse, and the gain identity $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$ together with its nonnegativity *and its strict positivity under any non-constant reward* (`gain_eq`, `gain_nonneg`, `gain_pos_of_nonconstant`, `kl_pos_of_ne`, `flat_no_gain`) — all **fully sorry-free**. This is exactly the formal content of the spread lens: the gain is a function of the *reward structure* (the range $R_{\max} - R_{\min}$ over $\text{supp}(q_0)$), and nothing in the hypotheses references a model class. Two **sorry-free** facts make the lens precise. (i) `gain_pos_of_nonconstant`: every non-degenerate verifiable reward contributes *strictly* positive gain, even on a single frozen model — this is why the high-spread content/intent rewards yield selection gains. (ii) `flat_no_gain`: a flat reward yields *zero* gain — this is why a factor with little reward spread (our paralinguistic negative) has nothing for training-free selection to exploit. The Lean project `proofs/tf1` may contain additional lemmas beyond those listed here; only the lemmas in Section A are load-bearing for this paper.

We are explicit about the boundary between the machine-checked identity and its quantitative companion. The qualitative identity $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$ and the strict lemmas above are machine-checked **sorry-free**. The *quantitative* spread bound $\mathcal{G} \leq \text{spread}^2/(8\beta)$ is the standard Donsker–Varadhan / Hoeffding statement (proved on paper in Lemma 1); we use it *qualitatively* — to say “small spread $\Rightarrow$ small headroom” — and we do **not** bill it as a novel or machine-checked ceiling. Throughout, $\mathcal{Z}$ is a **Fintype** (the inference-time action space), $q_0 : \mathcal{Z} \rightarrow \mathbb{R}$ is a strictly positive reference distribution with $\sum_z q_0(z) = 1$, $R : \mathcal{Z} \rightarrow \mathbb{R}$ is a reward, and $\beta > 0$ a temperature.

Verification status. To prevent any over-reading of “machine-checked”, we tabulate the exact status of each retained Lean result.

Lean lemma	Status	Undischarged input
<code>kl_nonneg</code>	sorry-free	—
<code>kl_pos_of_ne</code> (strict Gibbs)	sorry-free	—
<code>F_sub_eq_beta_mul_kl</code>	sorry-free	—
<code>tilting_optimal</code> (T1)	sorry-free	—
<code>qstar_eq_q0_of_const</code> (T3)	sorry-free	—
<code>gain_eq</code> (OSA-1a)	sorry-free	—
<code>gain_nonneg</code> (OSA-1b)	sorry-free	—
<code>gain_pos_of_nonconstant</code> (OSA-1b')	sorry-free	—
<code>flat_no_gain</code> (OSA-1c)	sorry-free	—

Every entry compiles with no axioms beyond mathlib’s and no **sorry**; the exact mathlib commit against which they compile is pinned in the `lake-manifest.json` committed with the `proofs/tf1` project, so the build is reproducible bit-for-bit. (“OSA” abbreviates *Optimization-Space Adequacy*; only the OSA-1 single-block lemmas are used in this paper.) The one quantitative statement we invoke — the $\text{spread}^2/(8\beta)$ ceiling — is **not** a Lean result: it is the standard Hoeffding bound, proved on paper in Lemma 1 and used only qualitatively.

Tilting primitives and the Gibbs inequality. The exponentially tilted optimum q^* , its partition function Z , and the KL-regularized objective $F(\cdot)$ are defined exactly as in the main text. We reproduce the Lean signatures verbatim.

```

noncomputable def Zpart (q0 R : Z → ℝ) (β : ℝ) : ℝ := ∑ w, q0 w * Real.exp (R w / β)
noncomputable def qstar (q0 R : Z → ℝ) (β : ℝ) (z : Z) : ℝ :=
  q0 z * Real.exp (R z / β) / Zpart q0 R β
noncomputable def F (q0 R : Z → ℝ) (β : ℝ) (q : Z → ℝ) : ℝ :=
  (∑ z, q z * R z) - β * ∑ z, q z * Real.log (q z / q0 z)

```

```

theorem kl_nonneg {p r : Z → ℝ} (hp : ∀ z, 0 ≤ p z) (hr : ∀ z, 0 < r z)
  (hpsum : ∑ z, p z = 1) (hrsum : ∑ z, r z = 1) :
  0 ≤ ∑ z, p z * Real.log (p z / r z)

```

`kl_nonneg` is Gibbs' inequality: the pointwise estimate $p(z) - r(z) \leq p(z) \log(p(z)/r(z))$ (trivial when $p(z) = 0$; otherwise $\log x \leq x - 1$ with $x = r(z)/p(z)$ via `Real.log_le_sub_one_of_pos`) is summed only over $\text{supp}(p)$. On that index set $\sum_{z \in \text{supp}(p)} p(z) = 1$, while $\sum_{z \in \text{supp}(p)} r(z) \leq \sum_z r(z) = 1$ (the partial sum over the support is $\leq$ the full sum, since $r \geq 0$); this gives $\sum_{z \in \text{supp}(p)} (p(z) - r(z)) \geq 1 - 1 = 0$, hence $\text{KL}(p \| r) \geq 0$. The one-sided bound “ $\leq$ the full sum” is sufficient.

We additionally prove the *strict* form, `sorry-free`:

```

theorem kl_pos_of_ne {p r : Z → ℝ} (hp : ∀ z, 0 < p z) (hr : ∀ z, 0 < r z)
  (hpsum : ∑ z, p z = 1) (hrsum : ∑ z, r z = 1) (hne : ∃ z, p z ≠ r z) :
  0 < ∑ z, p z * Real.log (p z / r z)

```

`kl_pos_of_ne` upgrades Gibbs to its equality case: $\text{KL}(p \| r) = 0 \iff p = r$, so $p \neq r \Rightarrow \text{KL}(p \| r) > 0$. The pointwise slack $p(z) \log(p(z)/r(z)) - (p(z) - r(z)) \geq 0$ is *strictly* positive at any z with $p(z) > 0$ and $p(z) \neq r(z)$ (strict form of $\log x < x - 1$ for $x \neq 1$, via `Real.log_lt_sub_one_of_ne`); the hypothesis $\exists z, p(z) \neq r(z)$ (with p strictly positive everywhere) supplies such a z , so summing the nonnegative slacks — at least one strictly positive — gives a strictly positive total. This strict inequality is the engine of the strict gain lemma below: it is exactly why a non-constant reward yields *strictly* positive gain rather than merely nonnegative gain.

Master identity and T1. The decomposition $F(q^*) - F(q) = \beta \text{KL}(q \| q^*)$ is the engine of optimality.

```

theorem F_sub_eq_beta_mul_kl [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  {q : Z → ℝ} (hq : ∀ z, 0 ≤ q z) (hqsum : ∑ z, q z = 1) :
  F q0 R β (qstar q0 R β) - F q0 R β q = β * ∑ z, q z * Real.log (q z / qstar q0 R β z)

```

```

theorem tilting_optimal [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  {q : Z → ℝ} (hq : ∀ z, 0 ≤ q z) (hqsum : ∑ z, q z = 1) :
  F q0 R β q ≤ F q0 R β (qstar q0 R β)

```

T1 (`tilting_optimal`) is then immediate: $F(q^*) - F(q) = \beta \text{KL}(q \| q^*) \geq 0$ by `kl_nonneg` and $\beta > 0$.

T3 — flat reward, no tilt.

```

theorem qstar_eq_q0_of_const [Nonempty Z] {q0 : Z → ℝ} {β c : ℝ}
  (hq0sum : ∑ z, q0 z = 1) {R : Z → ℝ} (hR : ∀ z, R z = c) :
  ∀ z, qstar q0 R β z = q0 z

```

When $R(z) \equiv c$, $Z = \exp(c/\beta)$ collapses and $q^*(z) = q_0(z) \exp(c/\beta) / \exp(c/\beta) = q_0(z)$: a degenerate action space with zero reward spread admits no steering.

The gain identity (TfmlProofs.OptSpace). We define the optimization gain $\mathcal{G} := F(q^*) - F(q_0)$ and prove its structural properties. Every lemma below compiles `sorry`-free.

```

noncomputable def gain (q0 R : Z → ℝ) (β : ℝ) : ℝ :=
  F q0 R β (qstar q0 R β) - F q0 R β q0

theorem gain_eq [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  (hq0sum : ∑ z, q0 z = 1) :
  gain q0 R β = β * Real.log (Zpart q0 R β) - ∑ z, q0 z * R z
theorem gain_nonneg [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  (hq0sum : ∑ z, q0 z = 1) : 0 ≤ gain q0 R β
theorem gain_pos_of_nonconstant [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  (hq0sum : ∑ z, q0 z = 1) (hR : ∃ z w, R z ≠ R w) :
  0 < gain q0 R β
theorem flat_no_gain [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
  (hq0sum : ∑ z, q0 z = 1) {c : ℝ} (hR : ∀ z, R z = c) : gain q0 R β = 0

```

`gain_eq` rewrites the gain as $\beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R]$ (**OSA-1a**); equivalently, read at $q = q_0$, the master identity gives the free-energy / Donsker–Varadhan form $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$. `gain_nonneg` (**OSA-1b**) is Jensen / Gibbs. `gain_pos_of_nonconstant` (**OSA-1b'**) is the strict companion: if the reward takes two distinct values, $R(z_1) \neq R(z_2)$, then $q^* \neq q_0$ (the tilt is non-trivial), so by the strict Gibbs lemma `kl_pos_of_ne` (applied with the strictly-positive argument $r = q^* > 0$) the identity $\mathcal{G} = \beta \text{KL}(q_0 \parallel q^*)$ is *strictly* positive. This is the formal statement of the spread lens at the single-block level: **any** non-degenerate verifiable reward yields strictly positive gain — crucially, *even on a single frozen model*, since nothing in the hypotheses references a model class. Conversely `flat_no_gain` (**OSA-1c**) recovers T3: a flat reward (e.g. a frozen embedding read-out that does not separate the labelled classes, hence a degenerate $\mathcal{Z}$ under that reward) affords *zero* improvement. Together OSA-1b'/OSA-1c make the dichotomy sharp and model-agnostic: gain is governed by whether the reward is non-constant. This is precisely the contrast the paper reports empirically — content/intent rewards have large spread (selection gains), the paralinguistic reward on this frozen encoder is near-flat (no significant gain).

B Extended Proof: the spread bound

We record, on paper, the standard analytic statement behind the spread lens: the reward spread caps the gain. This is the classical Donsker–Varadhan / Hoeffding bound; we present it for completeness and use it only qualitatively. It is *not* a Lean result and we do not bill it as one.

B.1 Hoeffding instance: the gain ceiling $\text{spread}^2/(8\beta)$

Lemma 1 (Bounded-reward gain ceiling). *Let $R : \mathcal{Z} \rightarrow [R_{\min}, R_{\max}]$ with reward spread $\text{spread} := R_{\max} - R_{\min}$, and let q_0 be any distribution on $\mathcal{Z}$ with $\mathbb{E}_{q_0}$ the corresponding expectation. Then for every $\beta > 0$,*

$$\mathcal{G}(q_0, R, \beta) = \beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R] \leq \frac{\text{spread}^2}{8\beta}.$$

Proof. Let $X := R(z)/\beta$ for $z \sim q_0$, supported in $[a, b]$ with $a = R_{\min}/\beta$, $b = R_{\max}/\beta$, so $b - a = \text{spread}/\beta$. Set $\varphi(\lambda) := \log \mathbb{E}[e^{\lambda X}]$; then $\varphi(0) = 0$, $\varphi'(0) = \mathbb{E}X$, and $\varphi''(\lambda) = \text{Var}_\lambda(X)$ under the tilted law $d\mu_\lambda \propto e^{\lambda x} d\mu$. Since $X \in [a, b]$ almost surely, the same holds μ_λ -a.s., and a random variable bounded in an interval of length $b - a$ has variance at most $(b - a)^2/4$ (extremal case: the two-point distribution at the endpoints). Hence $\varphi''(\lambda) \leq (b - a)^2/4$. By Taylor's theorem with the Lagrange remainder, for some ξ between 0 and λ ,

$$\varphi(\lambda) = \varphi(0) + \lambda\varphi'(0) + \frac{1}{2}\lambda^2\varphi''(\xi) \leq \lambda\mathbb{E}X + \frac{\lambda^2(b - a)^2}{8},$$

so $\psi(\lambda) := \varphi(\lambda) - \lambda \mathbb{E}X \leq \lambda^2(b-a)^2/8$, which is Hoeffding’s lemma. Taking $\lambda = 1$,

$$\log \mathbb{E}[e^X] - \mathbb{E}[X] = \psi(1) \leq \frac{(b-a)^2}{8} = \frac{\text{spread}^2}{8\beta^2}.$$

Multiplying by $\beta > 0$ and substituting $X = R/\beta$ gives $\beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R] \leq \text{spread}^2/(8\beta)$. By `gain_eq` the left side is exactly $\mathcal{G}$. $\square$

Lemma 1 makes the scaling explicit: the inference-time selection gain is quadratically throttled by the reward spread, $\mathcal{G} = O(\text{spread}^2/\beta)$, so a flat or near-flat reward landscape (T3, **OSA-1c**) yields vanishing headroom regardless of candidate budget — the analytic counterpart of reward over-optimization saturation [gao, 2022]. Together with `gain_pos_of_nonconstant` (strict lower bound for any non-constant reward), it brackets the gain from both sides: it is the reward spread, not the model class, that sets the ceiling. We use this bound qualitatively only.

C Notation

Symbol	Meaning
$\mathcal{Z}$	inference-time action space (candidate cards; no weights change)
q_0	base reference policy q_0 (strictly positive, $\sum_z q_0(z) = 1$)
q	a search distribution q over $\mathcal{Z}$
q^*	Gibbs / tilted optimum $q^*(z) = q_0(z)e^{R(z)/\beta}/Z$
R	verifiable reward function $\mathcal{Z} \rightarrow \mathbb{R}$
β	temperature (> 0); KL-regularization strength
Z	partition function $\sum_w q_0(w)e^{R(w)/\beta}$
$F(q)$	KL-regularized objective $\mathbb{E}_q[R] - \beta \text{KL}(q \parallel q_0)$
$\text{KL}(p \parallel q)$	Kullback–Leibler divergence
$\mathbb{E}$	expectation
$\mathcal{G}$	optimization gain $F(q^*) - F(q_0) = \beta \text{KL}(q_0 \parallel q^*) \geq 0$
spread	reward spread $R_{\max} - R_{\min}$
N	number of candidate cards in verifiable selection (best-of- N / reranking)

D Verifiable Reward Functions

The reward callables used as verifiable signals R are exposed by the shared library `speechrl_common.rl`, and are deliberately deterministic and contamination-resistant [tul, 2024]. Heavy dependencies (`jiwer`, `scikit-learn`) are imported lazily inside each function so the package and its smoke tests import without the full stack. In our experiments all selection is performed by a *frozen bi-encoder* ranking candidate cards by cosine under one of these rewards; no weights are updated.

Content / ASR (`rl.reward`). `wer(reference, hypothesis)` returns the word error rate after normalization; `asr_reward` returns $\max(0, 1 - \text{WER}) \in [0, 1]$, a bounded reward whose spread directly governs the gain ceiling of Lemma 1; `exact_match_reward` returns $\mathbf{1}[\text{normalized strings equal}] \in \{0, 1\}$ for closed-form tasks (intent, tool selection, keyword). These content/intent rewards are the ones for which our experiments realize a large, non-degenerate spread over the candidate cards — consistent with the frozen bi-encoder card-selection gains reported in the main text (MInDS-14 tool/intent, SLURP intent).

Retrieval / embedding (`rl.embedding_metrics`). `recall_at_k(query, gallery, labels_q, labels_g, k)` and `mean_reciprocal_rank(...)` score the frozen model’s pooled embeddings against a labelled gallery under cosine similarity (`cosine_sim_matrix` with ℓ_2 normalization); `retrieval_reward(..., k)` packages `recall@k` as a $[0, 1]$ reward. We flag honestly that for the *speaker* factor this reward is near-degenerate on the frozen omni embedding: the content read-out carries near-chance speaker identity

(mean accuracy 0.033, $3.0\times$ chance on a 91-way task; dev-selected layer, 3 seeds; committed artifact `_repro/paralinguistic_negative_probe.json`). This is a near-flat landscape under which Lemma 1 predicts vanishing gain — there is little paralinguistic spread for training-free selection to exploit.

Probe accuracy (`rl.probe`). `linear_probe_accuracy` / `knn_probe_accuracy` fit a frozen-feature classifier (held-out split) to score readability of the emotion (SER) and language/intent factors from the embeddings; `probe_reward(..., kind)` returns the held-out accuracy as the reward. For the *emotion* factor the frozen read-out is only modestly above chance (accuracy 0.405, $2.4\times$ chance on a 6-way CREMA-D task; dev-selected layer, 3 seeds; same committed artifact `_repro/paralinguistic_negative_probe.json`), and training-free pool-selection over this encoder yields *no significant gain*: a multi-seed re-run with dev-selected pooling layer (5 seeds: 42, 7, 123, 2024, 31337) gives mean $\Delta = +0.037$ with a 95% across-seed *t*-CI of $[-0.043, +0.116]$, which spans 0 — not significant (committed artifact `_repro/emotion_pool_paired_v2.json`).¹ The previously-reported single-seed $+0.097$ was the best seed under oracle test-layer selection (a selection-leak / winner’s-curse artifact). We scope this negative precisely: it is established on *one* frozen content bi-encoder (omni-embed) via a linear/kNN probe and pool-selection over its embedding space, and it says only that training-free selection over *this* encoder has little paralinguistic spread to exploit. Auxiliary metrics (`macro_f1`, `bleu`, `chrf`, `eer`) in `rl.metrics` are used for evaluation only, not as RL rewards, to avoid coupling the reward and the test statistic [ska, 2022].

References

- We Need to Talk about Standard Splits. <https://aclanthology.org/P19-1267/>.
- Task Contamination: Language Models May Not Be Few-Shot Anymore. <https://ojs.aaai.org/index.php/AAAI/article/view/29808>.
- Attentive Statistics Pooling for Deep Speaker Embedding. *arXiv preprint arXiv:1803.10963*, 2018. URL <https://arxiv.org/abs/1803.10963>.
- ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN. *arXiv preprint arXiv:2005.07143*, 2020. URL <https://arxiv.org/abs/2005.07143>.
- Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere. *arXiv preprint arXiv:2005.10242*, 2020. URL <https://arxiv.org/abs/2005.10242>.
- What Should Not Be Contrastive in Contrastive Learning. *arXiv preprint arXiv:2008.05659*, 2020. URL <https://arxiv.org/abs/2008.05659>.
- Layer-wise Analysis of a Self-supervised Speech Representation Model. *arXiv preprint arXiv:2107.04734*, 2021. URL <https://arxiv.org/abs/2107.04734>.
- SUPERB: Speech processing Universal PERFORMANCE Benchmark. *arXiv preprint arXiv:2105.01051*, 2021. URL <https://arxiv.org/abs/2105.01051>.
- Scaling Laws for Reward Model Overoptimization. *arXiv preprint arXiv:2210.10760*, 2022. URL <https://arxiv.org/abs/2210.10760>.
- RL with KL penalties is better viewed as Bayesian inference. *arXiv preprint arXiv:2205.11275*, 2022. URL <https://arxiv.org/abs/2205.11275>.
- Rethinking the Role of Demonstrations: What Makes In-Context Learning Work? *arXiv preprint arXiv:2202.12837*, 2022. URL <https://arxiv.org/abs/2202.12837>.
- Defining and Characterizing Reward Hacking. *arXiv preprint arXiv:2209.13085*, 2022. URL <https://arxiv.org/abs/2209.13085>.

¹The per-seed paired-delta bootstrap excludes 0 in 2/5 seeds; this vote-count is reported only for completeness and is not the headline statistic.

One Embedder, Any Task: Instruction-Finetuned Text Embeddings (INSTRUCTOR). *arXiv preprint arXiv:2212.09741*, 2022. URL <https://arxiv.org/abs/2212.09741>.

Self-Consistency Improves Chain of Thought Reasoning in Language Models. *arXiv preprint arXiv:2203.11171*, 2022. URL <https://arxiv.org/abs/2203.11171>.

Dynamic-SUPERB: A Dynamic, Collaborative, Comprehensive Instruction-Tuning Benchmark for Speech. *arXiv preprint arXiv:2309.09510*, 2023. URL <https://arxiv.org/abs/2309.09510>.

Generative Agents: Interactive Simulacra of Human Behavior. *arXiv preprint arXiv:2304.03442*, 2023. URL <https://arxiv.org/abs/2304.03442>.

Exploring In-Context Learning of Textless Speech Language Model for Speech Classification Tasks. *arXiv preprint arXiv:2310.12477*, 2023. URL <https://arxiv.org/abs/2310.12477>.

MemGPT: Towards LLMs as Operating Systems. *arXiv preprint arXiv:2310.08560*, 2023. URL <https://arxiv.org/abs/2310.08560>.

Controlled Decoding from Language Models. *arXiv preprint arXiv:2310.17022*, 2023. URL <https://arxiv.org/abs/2310.17022>.

SALMONN: Towards Generic Hearing Abilities for Large Language Models. *arXiv preprint arXiv:2310.13289*, 2023. URL <https://arxiv.org/abs/2310.13289>.

Function Vectors in Large Language Models. *arXiv preprint arXiv:2310.15213*, 2023. URL <https://arxiv.org/abs/2310.15213>.

Voyager: An Open-Ended Embodied Agent with Large Language Models. *arXiv preprint arXiv:2305.16291*, 2023. URL <https://arxiv.org/abs/2305.16291>.

Larger language models do in-context learning differently. *arXiv preprint arXiv:2303.03846*, 2023. URL <https://arxiv.org/abs/2303.03846>.

Theoretical guarantees on the best-of-n alignment policy. *arXiv preprint arXiv:2401.01879*, 2024. URL <https://arxiv.org/abs/2401.01879>.

Making Text Embedders Few-Shot Learners (bge-en-icl). *arXiv preprint arXiv:2409.15700*, 2024. URL <https://arxiv.org/abs/2409.15700>.

Benchmark Data Contamination of Large Language Models: A Survey. *arXiv preprint arXiv:2406.04244*, 2024. URL <https://arxiv.org/abs/2406.04244>.

Frozen Large Language Models Can Perceive Paralinguistic Aspects of Speech. *arXiv preprint arXiv:2410.01162*, 2024. URL <https://arxiv.org/abs/2410.01162>.

BoNBoN Alignment for LLMs and the Sweetness of Best-of-n Sampling. *arXiv preprint arXiv:2406.00832*, 2024. URL <https://arxiv.org/abs/2406.00832>.

Regularized Best-of-N Sampling with Minimum Bayes Risk Objective for LM Alignment. *arXiv preprint arXiv:2404.01054*, 2024. URL <https://arxiv.org/abs/2404.01054>.

Evaluating Very Long-Term Conversational Memory of LLM Agents (LoCoMo). *arXiv preprint arXiv:2402.17753*, 2024. URL <https://arxiv.org/abs/2402.17753>.

LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. *arXiv preprint arXiv:2410.10813*, 2024. URL <https://arxiv.org/abs/2410.10813>.

NV-Embed: Improved Techniques for Training LLMs as Generalist Embedding Models. *arXiv preprint arXiv:2405.17428*, 2024. URL <https://arxiv.org/abs/2405.17428>.

Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge. *arXiv preprint arXiv:2406.07791*, 2024. URL <https://arxiv.org/abs/2406.07791>.

Qwen2-Audio Technical Report. *arXiv preprint arXiv:2407.10759*, 2024. URL <https://arxiv.org/abs/2407.10759>.

Self-Preference Bias in LLM-as-a-Judge. *arXiv preprint arXiv:2410.21819*, 2024. URL <https://arxiv.org/abs/2410.21819>.

Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters. *arXiv preprint arXiv:2408.03314*, 2024. URL <https://arxiv.org/abs/2408.03314>.

Tulu 3: Pushing Frontiers in Open Language Model Post-Training. *arXiv preprint arXiv:2411.15124*, 2024. URL <https://arxiv.org/abs/2411.15124>.

VoiceBench: Benchmarking LLM-Based Voice Assistants. *arXiv preprint arXiv:2410.17196*, 2024. URL <https://arxiv.org/abs/2410.17196>.

Asymptotics of Language Model Alignment. *arXiv preprint arXiv:2404.01730*, 2024. URL <https://arxiv.org/abs/2404.01730>.

Best-of-N through the Smoothing Lens: KL Divergence and Regret Analysis. *arXiv preprint arXiv:2507.05913*, 2025. URL <https://arxiv.org/abs/2507.05913>.

Theoretical Guarantees for Minimum Bayes Risk Decoding. *arXiv preprint arXiv:2502.12685*, 2025. URL <https://arxiv.org/abs/2502.12685>.

Inference-Time Reward Hacking in Large Language Models (HedgeTune, Best-of-Poisson). *arXiv preprint arXiv:2506.19248*, 2025. URL <https://arxiv.org/abs/2506.19248>.

Test-Time Preference Optimization: On-the-Fly Alignment via Iterative Textual Feedback. *arXiv preprint arXiv:2501.12895*, 2025. URL <https://arxiv.org/abs/2501.12895>.

Do Audio LLMs Really LISTEN, or Just Transcribe? Measuring Lexical vs. Acoustic Emotion Cues Reliance. *arXiv preprint arXiv:2510.10444*, 2025. URL <https://arxiv.org/abs/2510.10444>.

Re-evaluating Minimum Bayes Risk Decoding for Automatic Speech Recognition. *arXiv preprint arXiv:2510.19471*, 2025. URL <https://arxiv.org/abs/2510.19471>.

Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory. *arXiv preprint arXiv:2504.19413*, 2025a. URL <https://arxiv.org/abs/2504.19413>.

Memory-R1: Enhancing LLM Agents to Manage and Utilize Memories via Reinforcement Learning. *arXiv preprint arXiv:2508.19828*, 2025b. URL <https://arxiv.org/abs/2508.19828>.

MiMo-Audio: Audio Language Models are Few-Shot Learners. *arXiv preprint arXiv:2512.23808*, 2025. URL <https://arxiv.org/abs/2512.23808>.

Omni-Embed-Nemotron: A Unified Multimodal Retrieval Model for Text, Image, Audio, and Video. *arXiv preprint arXiv:2510.03458*, 2025. URL <https://arxiv.org/abs/2510.03458>.

Qwen2.5-Omni Technical Report. *arXiv preprint arXiv:2503.20215*, 2025a. URL <https://arxiv.org/abs/2503.20215>.

Qwen3-Omni Technical Report. *arXiv preprint arXiv:2509.17765*, 2025b. URL <https://arxiv.org/abs/2509.17765>.

RLVR Implicitly Incentivizes Correct Reasoning in Base LLMs. *arXiv preprint arXiv:2506.14245*, 2025. URL <https://arxiv.org/abs/2506.14245>.

Few-shot Personalization via In-Context Learning for Speech Emotion Recognition based on Speech-LM. *arXiv preprint arXiv:2509.08344*, 2025. URL <https://arxiv.org/abs/2509.08344>.

A Large-Scale Probing Analysis of Speaker-Specific Attributes in SSL Speech Representations. *arXiv preprint arXiv:2501.05310*, 2025. URL <https://arxiv.org/abs/2501.05310>.

Spurious Rewards: Rethinking Training Signals in RLVR. *arXiv preprint arXiv:2506.10947*, 2025. URL <https://arxiv.org/abs/2506.10947>.

Task Vectors in In-Context Learning: Emergence, Formation, and Benefits. *arXiv preprint arXiv:2501.09240*, 2025. URL <https://arxiv.org/abs/2501.09240>.

URO-Bench: Towards Comprehensive Evaluation for End-to-End Spoken Dialogue Models. *arXiv preprint arXiv:2502.17810*, 2025. URL <https://arxiv.org/abs/2502.17810>.

Soft Best-of-n Sampling for Model Alignment. *arXiv preprint arXiv:2505.03156*, 2025. URL <https://arxiv.org/abs/2505.03156>.

TTRL: Test-Time Reinforcement Learning. *arXiv preprint arXiv:2504.16084*, 2025. URL <https://arxiv.org/abs/2504.16084>.

ALICE: A Multifaceted Evaluation Framework of Large Audio-Language Models In-Context Learning Ability. *arXiv preprint arXiv:2603.20433*, 2026. URL <https://arxiv.org/abs/2603.20433>.

A-MBER: Affective Memory Benchmark for Emotion Recognition. *arXiv preprint arXiv:2604.07017*, 2026. URL <https://arxiv.org/abs/2604.07017>.

Anatomy of Agentic Memory: Taxonomy and Empirical Analysis of Evaluation and System Limitations. *arXiv preprint arXiv:2602.19320*, 2026. URL <https://arxiv.org/abs/2602.19320>.

Discovering and Causally Validating Emotion-Sensitive Neurons in Large Audio-Language Models. *arXiv preprint arXiv:2601.03115*, 2026. URL <https://arxiv.org/abs/2601.03115>.

Just-In-Time Reinforcement Learning: Continual Learning in LLM Agents Without Gradient Updates (ICML 2026 Spotlight). *arXiv preprint arXiv:2601.18510*, 2026. URL <https://arxiv.org/abs/2601.18510>.

New Skills or Sharper Primitives? A Probabilistic Perspective on the Emergence of Reasoning in RLVR. *arXiv preprint arXiv:2602.08281*, 2026. URL <https://arxiv.org/abs/2602.08281>.

SoK: Agentic Skills - Beyond Tool Use in LLM Agents. *arXiv preprint arXiv:2602.20867*, 2026. URL <https://arxiv.org/abs/2602.20867>.

Speaker Verification with Speech-Aware LLMs: Evaluation and Augmentation. *arXiv preprint arXiv:2603.10827*, 2026. URL <https://arxiv.org/abs/2603.10827>.

Do Audio LLMs Listen or Read? Analyzing and Mitigating Paralinguistic Failures with VoxParadox. *arXiv preprint arXiv:2605.27772*, 2026. URL <https://arxiv.org/abs/2605.27772>.